The first finite-copy correction in two-parameter qubit estimation

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Joint measurements of a finite number of quantum-state copies must balance information about incompatible parameters. The correction to infinite-copy precision quantifies the cost of this finite measurement resource. We determine the first finite-copy correction to the attainable locally unbiased risk for the two-component qubit model at the maximally mixed state. If an arbitrary joint measurement acts on N copies and the total mean squared error is multiplied by N, its optimum is $2 + \sqrt{8/\pi}\, N^{-1/2} + o(N^{-1/2})$. The central result is that the common convex roof of the two transverse spin variances at $I/(2j+1)$ equals $j/2 + o(j)$. We prove the upper bound using Gaussian columns with a position-dependent width. Their populations integrate to one exactly, so a finite phase average yields a complete positive operator-valued measure for every finite spin. A nonnegative energy decomposition controls the entire construction, including both poles and Gaussian tails. A sum-of-squares certificate supplies the matching lower bound for all measurements. The construction has an explicit width parameter and an exactly locally unbiased score estimator for every finite N. Consequently, the strict finite-copy Nagaoka–Hayashi gap present for $N \geq 10$ is $o(N^{-1/2})$.

I. INTRODUCTION

Joint measurements on finitely many copies of a quantum state must balance information about incompatible observables. Even when the infinite-copy precision is known, its first finite-copy correction can depend on whether low-variance measurement effects can be assembled into an exactly complete measurement. We study this question for the two Cartesian components of a qubit at the maximally mixed state.

General conic formulations characterize the attainable local risk and allow reconstruction of complete measurements [1]. Schur–Weyl reduction makes finite-copy problems considerably smaller [2]. These results provide a finite-dimensional optimization framework; convergence of a measurement grid at fixed N does not determine the rate at which C_N approaches its infinite-copy limit. The present problem asks for that rate and a complete measurement family attaining it.

Planar spin squeezing and Gaussian descriptions of low-variance spin states are established tools [4, 5]. In particular, the fixed-squeezing asymptotics of [5, Theorem 4], after rotation and optimization, give the local cost $j|x|$ at a fixed nonzero latitude x. This suggests the formal average $j\int_{-1}^{1}|x|\,dx/2 = j/2$. The missing condition is an exact decomposition of I/d: independently chosen low-variance states need not have the required average populations. Our construction imposes that condition column by column and controls its average cost globally. Thus the new asymptotic result does not require a new Schur reduction, convex-roof identity, or local squeezing principle.

We determine the optimal normalized risk to its first correction, $C_N = 2 + \sqrt{8/\pi}\, N^{-1/2} + o(N^{-1/2})$, and provide a single explicit family attaining it. Every finite member is a complete POVM with an exactly locally unbiased estimator. The strict finite-copy Nagaoka–Hayashi (NH) gap for $N \geq 10$ is therefore subleading on the $N^{-1/2}$ scale.

Section II defines the statistical problem and its spin reduction. Section III states the main result, and Sec. IV specifies the measurement and estimator. Section V discusses finite-copy comparisons. The appendixes supply all technical proofs.

II. MODEL AND ATTAINABLE PRECISION

Consider the full-rank local qubit model

$$\rho_{x,y} = \frac{1}{2}(I + x\sigma_x + y\sigma_y), \qquad x^2 + y^2 < 1, \tag{1}$$

at $(x, y) = (0, 0)$. The parameters are Cartesian components of the density matrix. For each positive integer N, a single arbitrary joint POVM is applied to $\rho_{x,y}^{\otimes N}$. Its outcome may be discrete or continuous. We require the estimator $\widehat{\theta} = (\widehat{x}, \widehat{y})$ to satisfy

$$\mathbb{E}_0\widehat{\theta}_a = 0, \qquad \partial_{\theta_b}\mathbb{E}_\theta\widehat{\theta}_a\Big|_0 = \delta_{ab}, \qquad \theta = (x, y), \tag{2}$$

and define

$$C_N = N \inf_{M,\widehat{\theta}} \mathbb{E}_0(\widehat{x}^2 + \widehat{y}^2). \tag{3}$$

Both the measurement and estimator may depend on the reference point. There is no restriction on ancillary systems, classical randomization, the number of outcomes, or the rank of the effects.

Write J_x, J_y, J_z for the spin-j angular momentum operators and $d = 2j + 1$. The relevant common variance convex roof is

$$V_j = \min_{\sum_a p_a|\psi_a\rangle\langle\psi_a| = I/d} \sum_a p_a\big(\mathrm{Var}_{\psi_a} J_x + \mathrm{Var}_{\psi_a} J_y\big). \tag{4}$$

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Both variances use the same decomposition. Such common convex roofs are part of the general framework of uncertainty relations based on pure state decompositions [3].

Let $\mathcal{J}_N = \{j_{\min}, j_{\min}+1, \ldots, N/2\}$, with $j_{\min} = 0$ for even N and $1/2$ for odd N. The Schur decomposition and its probabilities at the reference point are

$$\begin{aligned}(\mathbb{C}^2)^{\otimes N} &= \bigoplus_{j\in\mathcal{J}_N} \mathcal{H}_j \otimes \mathbb{C}^{\nu_j},\\ \nu_j &= \binom{N}{N/2-j} - \binom{N}{N/2-j-1},\\ P_{N,j} &= \frac{(2j+1)\nu_j}{2^N}.\end{aligned} \tag{5}$$

An out-of-range binomial coefficient is zero. Conditional on j, the reference state and its two tangent operators are

$$Q_j = \frac{I}{d}, \qquad \dot{Q}_{j,x} = \frac{2J_x}{d}, \qquad \dot{Q}_{j,y} = \frac{2J_y}{d}. \tag{6}$$

The probabilities $P_{N,j}$ have zero first derivatives in both parameters. Write $\mathbb{E}_N$ for expectation under $P_{N,j}$.

Lemma 1. *If T_j is the maximum classical Fisher trace in the spin-j model (6), then*

$$\begin{aligned}T_N &= \sum_j P_{N,j} T_j, \qquad C_N = \frac{4N}{T_N},\\ T_j &= \frac{8}{3} j(j+1) - 4V_j.\end{aligned} \tag{7}$$

These identities hold for the full class of POVMs, with no additional sample-size limit.

The proof, including the treatment of arbitrary complex and continuous-outcome POVMs, is given in Appendix A. The maximum is attained, so the optimization defines an attainable local risk for each finite N.

III. OPTIMAL FINITE-COPY CORRECTION

Theorem 2. *As $j \to \infty$ through integer or half-integer spins,*

$$V_j = \frac{j}{2} + o(j). \tag{8}$$

For the model (1), optimized over all joint POVMs and all estimators satisfying (2),

$$\boxed{C_N = 2 + \sqrt{\frac{8}{\pi}}\, N^{-1/2} + o(N^{-1/2}).} \tag{9}$$

The Gaussian-column measurement in Section IV, with

$$\epsilon_j = \frac{1}{\log(e + j + 1/2)}, \qquad j > 0, \tag{10}$$

and the estimator in (19) are well-defined, complete, and exactly locally unbiased for every positive integer N. Their risk has the same expansion (9).

The proof separates completeness from the local squeezing calculation. Appendix B gives a sum-of-squares certificate for all POVMs, with lower bound $L_j = j/2 + o(j)$. Appendix C proves that the exactly complete Gaussian family has average variance $W_j^{\mathrm{G}} = j/2 + o(j)$. Thus $L_j \le V_j \le W_j^{\mathrm{G}}$ determines the spin asymptotic.

The Schur mean spin satisfies $\mathbb{E}_N j/\sqrt{N} \to \sqrt{2/\pi}$. For any fixed spin sequence $r_j = o(j)$, Appendix D proves $\mathbb{E}_N r_j = o(\sqrt{N})$ by a direct finite-head and large-spin split. This supplies the uniformity needed to obtain the $N^{-1/2}$ correction from the spin result. In particular, the proof covers both parities of N and includes all small-spin sectors.

IV. COMPLETE MEASUREMENT

Fix $j > 0$ and set $J = j + 1/2$. Initially let $b \in C^2([-1,1])$ be positive, and define

$$\begin{aligned}a(x) &= (1-x^2)b(x), \qquad S(s) = Ja(s/J),\\ v_s(t) &= (2\pi S(s))^{-1/4} \exp\left[-\frac{(t-s)^2}{4S(s)}\right],\\ &\qquad |s| \le j.\end{aligned} \tag{11}$$

Use the discrete columns $v_m(t)$, $m = -j, \ldots, j$. With $c_m = \sqrt{(j-m)(j+m+1)}$, put

$$\begin{aligned}s_j(t) &= \sum_m v_m(t)^2,\\ h_j(t) &= \sum_{m=-j}^{j-1} c_m v_m(t) v_{m+1}(t).\end{aligned} \tag{12}$$

For an integer $L \ge \max(d, 3)$ set $\phi_\ell = 2\pi\ell/L$, $\ell = 0, \ldots, L-1$, and define the POVM density

$$\begin{aligned}E_j(dt, \ell) &= \frac{1}{L} |v_{t,\ell}\rangle\langle v_{t,\ell}|\, dt,\\ |v_{t,\ell}\rangle &= \sum_m v_m(t) e^{-im\phi_\ell} |m\rangle.\end{aligned} \tag{13}$$

Each $S(m) > 0$, since $|m| \le J - 1/2$. Every effect is positive, and $\int_{\mathbb{R}} v_m(t)^2\, dt = 1$. Averaging the phases eliminates all off-diagonal entries exactly; consequently

$$\sum_{\ell=0}^{L-1} \int_{\mathbb{R}} E_j(dt, \ell) = I. \tag{14}$$

No truncation or subsequent normalization is used. The associated decomposition of I/d has density $s_j(t)/(dL)$ and normalized vectors $v_{t,\ell}/\sqrt{s_j(t)}$.

Let $W_j[b]$ be its average planar variance and put $A_m = j(j+1) - m^2$. Phase rotations preserve the sum of the two variances, so

$$W_j[b] = \frac{1}{d} \int_{\mathbb{R}} \left[\sum_m A_m v_m(t)^2 - \frac{h_j(t)^2}{s_j(t)}\right] dt. \tag{15}$$

For clarity, the measurement of Theorem 2 uses the following fully specified width:

$$S_m = \frac{J^2 - m^2}{2\sqrt{m^2 + \epsilon_j^2 J^2}}, \qquad J = j + \frac{1}{2}, \tag{16}$$

and one may set $L = \max(2j+1, 3)$. First measure the Schur label j. For $j > 0$ apply (13), tensor the effects with the multiplicity identity, and for $j = 0$ use its identity effect. At outcome (j, t, ℓ), the conditional score is

$$u(j,t,\ell) = \frac{2h_j(t)}{s_j(t)} \begin{pmatrix} \cos\phi_\ell \\ \sin\phi_\ell \end{pmatrix}. \tag{17}$$

Indeed $\langle v_{t,\ell}|J_+|v_{t,\ell}\rangle = e^{i\phi_\ell} h_j(t)$ with the phase convention in (13). Let $W_j^{\mathrm{G}} = W_j[b_{\epsilon_j}]$, $W_0^{\mathrm{G}} = 0$, and set

$$T_N^{\mathrm{G}} = 2N - 4\mathbb{E}_N W_j^{\mathrm{G}}. \tag{18}$$

All Gaussian columns are strictly positive, so $h_j(t) > 0$ for $j > 0$. Hence $T_N^{\mathrm{G}} > 0$ for every $N \geq 1$ and the full Fisher matrix is $(T_N^{\mathrm{G}}/2) I_2$. The estimator is

$$\begin{aligned} \widehat{\theta}(j,t,\ell) &= \frac{4h_j(t)}{T_N^{\mathrm{G}} s_j(t)} \begin{pmatrix} \cos\phi_\ell \\ \sin\phi_\ell \end{pmatrix}, \\ \widehat{\theta}(0) &= 0. \end{aligned} \tag{19}$$

It uses the total Fisher information, common to all sectors. The score argument in Lemma 1 proves exact local unbiasedness and risk $C_N^{\mathrm{G}} = 4N/T_N^{\mathrm{G}}$.

V. FINITE-COPY COMPARISON

The capped quantities in (B2) give the Nagaoka–Hayashi value for the present model:

$$C_N^{\mathrm{NH}} = \frac{N}{\sum_j P_{N,j} S_j/(2j+1)} = \frac{4N}{2N - 4\mathbb{E}_N L_j}. \tag{20}$$

For completeness, the reduction and a proof of strict separation are given in Appendix E. They imply $C_N > C_N^{\mathrm{NH}}$ for $N \geq 10$. Equations (B3) and (D4) now yield

Corollary 3. *For every $N \geq 10$, $C_N - C_N^{\mathrm{NH}} > 0$, whereas*

$$C_N - C_N^{\mathrm{NH}} = o(N^{-1/2}) \qquad (N \to \infty). \tag{21}$$

The obstruction is an exact population constraint: the seeds saturating the capped certificate cannot resolve the identity in large enough spin sectors. The Gaussian columns satisfy that constraint exactly while losing only $o(j)$ in average planar variance. Schur averaging turns this into an $o(\sqrt{N})$ Fisher loss and an $o(N^{-1/2})$ risk gap.

Figure 1 illustrates the construction by direct floating-point integration. At $j = 3000$ its variance ratio is $W_j^{\mathrm{G}}/j \simeq 0.507269852$; the largest computed column normalization residual is below 2×10^{-12}. These numerical checks illustrate the exact analytic construction and are not used in the proof. The logarithmic width rule is chosen for a simple uniform asymptotic guarantee and need not optimize small spins. For example, at $N = 400$, its risk is approximately 2.11276018, compared with the coherent-state risk 2.10872335 and the NH lower bound 2.07352470.

VI. CONCLUSIONS

The first finite-copy correction for two-component qubit estimation at the maximally mixed state is $\sqrt{8/\pi}$. The result follows from an exact identity decomposition whose common planar variance is $j/2 + o(j)$, together with a matching certificate valid for all measurements.

Gaussian columns with position-dependent widths enforce population completeness before the phase average. Their nonnegative energy decomposition controls the poles, the equatorial regularization, and the full Gaussian tails. The resulting score estimator is exactly locally unbiased for every finite N. The logarithmic width sequence is a simple asymptotic construction; optimizing its finite-copy performance is a separate question. The finite-dimensional NH obstruction survives as a strict inequality but contributes only $o(N^{-1/2})$ to the normalized risk.

DATA AVAILABILITY

The numerical data, illustration code, and Lean 4 proof sources are included in the accompanying reproduction package.

ACKNOWLEDGMENTS

[Acknowledgments to be added.]

Appendix A: Exact statistical reduction and seed representation

This appendix proves Lemma 1 and establishes the seed representation used in the lower bound.

Proof. The global reference state and tangents are $I/2^N$ and $2J_a^{\mathrm{tot}}/2^N$. They are block diagonal in (5) and scalar on each multiplicity space. For an arbitrary effect M_z, define $E_{z|j} = \nu_j^{-1} \operatorname{tr}_{\mathrm{mult}}(\Pi_j M_z \Pi_j)$. Discarding the label j reproduces the original local probabilities and derivatives. Keeping it is a statistical refinement and increases Fisher information in positive semidefinite order by Cauchy–Schwarz. Conversely, any conditional POVM lifts to $E_{z|j} \otimes I_{\nu_j}$. Since $\dot{P}_{N,j} = 0$, its Fisher matrix is the weighted sum of the conditional matrices.

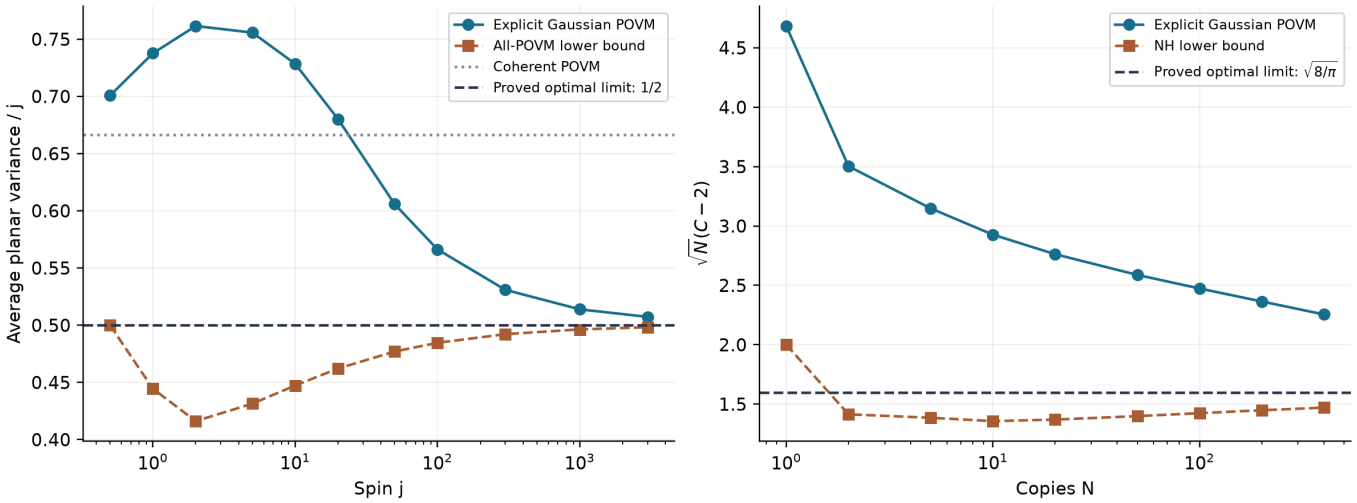


FIG. 1. Numerical illustration of the explicit Gaussian family, the NH lower bound, and the coherent-state benchmark. The Gaussian measurement uses (10); plotted values are floating-point diagnostics, not finite-spin optimality certificates. The limiting constants follow from the analytic proof.

For any measurement, let $u(z) = \nabla \log p_\theta(z)|_0$ and $F = \mathbb{E}_0[uu^\mathsf{T}]$. When F is invertible, the estimator $F^{-1}u$ satisfies (2) and has covariance F^{-1}. For every locally unbiased estimator,

$$\mathbb{E}_0\left[(\widehat{\theta} - F^{-1}u)(\widehat{\theta} - F^{-1}u)^\mathsf{T}\right] = \mathrm{Cov}_0(\widehat{\theta}) - F^{-1} \succeq 0.$$

If F is singular, the identity $\mathbb{E}_0[\widehat{\theta}u^\mathsf{T}] = I_2$ is impossible. Full rank of the reference state ensures that these score identities also apply to continuous outcomes, using the scalar trace measure as a dominating measure. In particular the scores are bounded. For positive definite F, $\mathrm{tr}\, F^{-1} \geq 4/\mathrm{tr}\, F$. Randomizing the measurement by rotations about z and retaining the rotation label gives $F = (\mathrm{tr}\, F/2)I_2$ without changing its trace. This proves the first two assertions.

Refining effects to rank one cannot decrease Fisher information. A pure state decomposition of I/d corresponds to the effects $E_a = dp_a|\psi_a\rangle\langle\psi_a|$. Its probability is p_a and its score is $2(\langle J_x\rangle_{\psi_a}, \langle J_y\rangle_{\psi_a})$. Completeness fixes

$$\sum_a p_a \langle J_x^2 + J_y^2\rangle_{\psi_a} = \frac{1}{d}\,\mathrm{tr}(J_x^2 + J_y^2) = \frac{2}{3}j(j+1).$$

Subtracting the average variance gives the last identity. □

For use in the lower bound, we give a real nonnegative seed form. In the J_z basis, indexed by $m = -j, \ldots, j$, set

$$c_m = \sqrt{(j-m)(j+m+1)}, \qquad n_a = \sum_m v_{a,m}^2,$$
$$h_a = \sum_{m=-j}^{j-1} c_m v_{a,m} v_{a,m+1}. \tag{A1}$$

Then

$$T_j = \frac{4}{d} \max_{v_a \geq 0,\ \sum_a v_{a,m}^2 = 1} \sum_a \frac{h_a^2}{n_a}. \tag{A2}$$

Indeed, for a complex rank-one vector w its trace contribution is $4|\langle w|J_+|w\rangle|^2/(d\|w\|^2)$. Replacing each coordinate by its modulus increases this quantity and preserves diagonal populations. Completeness is then restored exactly by the phase orbit

$$E_{a\ell} = \frac{1}{L}|v_{a\ell}\rangle\langle v_{a\ell}|, \qquad v_{a\ell,m} = v_{a,m}e^{-im\phi_\ell},$$
$$\phi_\ell = \frac{2\pi\ell}{L}, \qquad L \geq \max(d, 3). \tag{A3}$$

All nonzero frequencies $m - m'$ have absolute value below L, and their phase averages vanish. The first and second phase moments give zero score mean and an isotropic Fisher matrix. This argument also covers half-integer spins, since $m - m'$ is always an integer.

The maximum in (A2) is attained. To see this, put $p_{a,m} = v_{a,m}^2/n_a$ and $\mu_a = n_a/d$. These define a probability measure on the compact probability simplex with mean $(1/d, \ldots, 1/d)$. Its objective is the integral of the continuous function $4(\sum_m c_m\sqrt{p_m p_{m+1}})^2$. Weak compactness gives an optimizer. A supporting affine function for its concave envelope at the uniform vector, followed by Carathéodory's theorem on the contact set, reduces the support to at most d seeds. We will use attainment only when discussing strict finite-dimensional gaps.

Appendix B: A sum-of-squares lower bound

In this appendix index the d nodes by $i = 0, \ldots, d-1$ and the $d-1$ edges by $i = 0, \ldots, d-2$. Thus $c_i =$

$\sqrt{(i+1)(d-1-i)}$. For $j > 0$, let τ_j be the unique positive solution of

$$\sum_{i=0}^{d-2}(c_i - \tau_j)_+ = \tau_j. \tag{B1}$$

Write

$$\begin{aligned} a_i &= \min(c_i, \tau_j), \qquad S_j = \sum_i c_i a_i, \\ D_j &= \sum_i c_i^2 - S_j, \qquad L_j = \frac{D_j}{d}. \end{aligned} \tag{B2}$$

Set all four quantities to zero when $j = 0$.

Proposition 4. *For every spin, $V_j \geq L_j$. Moreover,*

$$L_j = \frac{j}{2} + o(j), \qquad 0 \leq L_j \leq \frac{j}{2} + \frac{1}{4}. \tag{B3}$$

Proof. The left side of (B1) minus its right side is continuous and strictly decreasing, positive at zero and negative at $\max_i c_i$. Define

$$r_0 = 0, \qquad r_{i+1} - r_i = c_i/a_i - 1.$$

Equation (B1) gives $r_{d-1} = 1$ and $0 \leq r_i \leq 1$. Put $a_{-1} = a_{d-1} = 0$ and $\lambda_i = (1-r_{i-1})a_{i-1}^2 + r_{i+1}a_i^2$, omitting absent boundary terms. If K has off-diagonal entries $K_{i,i+1} = c_i$, direct expansion gives, for every real v and t,

$$\begin{aligned} &v^{\mathsf{T}}(\operatorname{diag}\lambda + t^2 I - tK)v \\ &\quad = \sum_{i=0}^{d-1}\big[(1-r_i)(tv_i - a_i v_{i+1})^2 \\ &\qquad + r_i(tv_i - a_{i-1}v_{i-1})^2\big] \geq 0. \end{aligned} \tag{B4}$$

Here absent vector components are zero, and $a_i(1 - r_i + r_{i+1}) = c_i$. Also $\sum_i \lambda_i = S_j$. For a nonzero seed take $t = h_a/n_a$ in (B4). Then $h_a^2/n_a \leq \sum_i \lambda_i v_{a,i}^2$. Summing over the complete seed family yields $T_j \leq 4S_j/d$. Since $\sum_i c_i^2 = \frac{2}{3}j(j+1)d$, Lemma 1 proves $V_j \geq D_j/d$. Thus (B4) is itself a certificate for the entire POVM optimization.

For the asymptotic statement let $J = j + 1/2 = d/2$. Then $c_i^2 = J^2 - (i+1-J)^2$ and $\tau_j < J$. The cap equation implies

$$\tau_j^2 \leq D_j = \sum_i c_i(c_i - \tau_j)_+ \leq J\tau_j \leq J^2. \tag{B5}$$

If $\tau_j \leq (1-\eta)J$ on an unbounded subsequence for some $\eta > 0$, a fixed positive fraction of the central edges satisfy $c_i \geq (1 - \eta/2)J$. Their contribution to $\sum_i (c_i - \tau_j)_+$ is of order J^2, contradicting (B1). Hence $\tau_j/J \to 1$ and $D_j/J^2 \to 1$. Dividing (B5) by $d = 2J$ proves both claims. □

Appendix C: Global Gaussian energy estimates

The denominator in Eq. (15) couples all columns. The following bound removes it. Set

$$\begin{aligned} q(t) &= \sqrt{(J^2 - t^2)_+}, \qquad B_m = \sqrt{J^2 - m^2}, \\ k_m &= \frac{c_{m-1} + c_m}{2}, \end{aligned} \tag{C1}$$

where $c_{-j-1} = c_j = 0$. From $h^2/s \geq 2qh - q^2 s$,

$$\begin{aligned} W_j[b] &\leq U_j[b] \\ &:= \frac{1}{d}\int_{\mathbb{R}}\left[\sum_m (A_m + q^2)v_m^2 - 2qh_j\right]dt \\ &= \frac{1}{d}\left(\sum_m I_m + \sum_{m=-j}^{j-1} G_m + \sum_m R_m\right), \end{aligned} \tag{C2}$$

where every term is nonnegative:

$$\begin{aligned} I_m &= \int_{\mathbb{R}} (k_m - q(t))^2 v_m(t)^2 dt, \\ G_m &= c_m \int_{\mathbb{R}} q(t)(v_{m+1}(t) - v_m(t))^2 dt, \\ R_m &= \frac{(c_m - c_{m-1})^2}{4}. \end{aligned} \tag{C3}$$

To verify the second equality in (C2), use

$$\begin{aligned} h_j &= \sum_m k_m v_m^2 - \frac{1}{2}\sum_m c_m (v_{m+1} - v_m)^2, \\ A_m - k_m^2 &= R_m. \end{aligned}$$

Theorem 5. *For every fixed positive $b \in C^2([-1,1])$,*

$$\lim_{j\to\infty} \frac{U_j[b]}{j} = \frac{1}{2}\int_{-1}^{1}\left(x^2 b(x) + \frac{1}{4b(x)}\right)dx. \tag{C4}$$

For $b_\epsilon(x) = (2\sqrt{x^2 + \epsilon^2})^{-1}$ and the explicit sequence (10),

$$\frac{U_j[b_{\epsilon_j}]}{j} \longrightarrow \frac{1}{2}, \qquad W_j[b_{\epsilon_j}] = \frac{j}{2} + o(j). \tag{C5}$$

1. Uniform control at the poles

All constants denoted C_b in this subsection depend only on a positive lower bound for b and an upper bound for its C^2 norm. For $r(s) = J - |s| \geq 1/2$,

$$\begin{aligned} &Jr(s) \leq B(s)^2 := J^2 - s^2 \leq 2Jr(s), \\ &S(s) \asymp_b r(s), \qquad |S'(s)| \leq C_b, \\ &|S''(s)| \leq C_b/J. \end{aligned} \tag{C6}$$

The positive-part map is 1-Lipschitz. Therefore, for every real t, including $|t| \geq J$,

$$|q(t) - B_m| \leq \frac{|t^2 - m^2|}{B_m}. \tag{C7}$$

Under $v_m(t)^2dt$, $t-m$ is normal with variance $S_m = S(m)$. Taking its second and fourth moments gives

$$\int (q - B_m)^2 v_m^2 dt \le \frac{4m^2 S_m + 3S_m^2}{B_m^2} \le C_b J. \tag{C8}$$

Since $c_m = q(m + 1/2)$, the same inequality gives $|k_m - B_m|^2 \le CJ/r(m)$. Consequently, for every column,

$$0 \le I_m \le C_b J. \tag{C9}$$

For the gradient term write $Y = (t-s)/\sqrt{S(s)}$. Direct differentiation yields

$$\partial_s v_s = v_s \left[\frac{Y}{2\sqrt{S}} + \frac{S'}{4S}(Y^2 - 1) \right]. \tag{C10}$$

The square-root Hölder inequality and (C6) imply

$$\begin{aligned} q(s + \sqrt{S}Y) &\le B(s) + \sqrt{2J\sqrt{S}|Y| + SY^2} \\ &\le C_b \sqrt{Jr(s)}(1 + |Y|). \end{aligned} \tag{C11}$$

This bound also includes all tails crossing the poles. Gaussian moments in (C10) now give

$$\begin{aligned} &\int q(t)(\partial_s v_s(t))^2 dt \\ &\quad \le C_b \sqrt{Jr(s)} \left(\frac{1}{S(s)} + \frac{S'(s)^2}{S(s)^2} \right) \\ &\quad \le C_b \sqrt{\frac{J}{r(s)}}. \end{aligned} \tag{C12}$$

For $s \in [m, m+1]$, $c_m \le C\sqrt{Jr(s)}$. Integral Cauchy–Schwarz in s therefore yields

$$\begin{aligned} 0 \le G_m &\le c_m \int_m^{m+1} \int q(t)(\partial_s v_s(t))^2 dt\, ds \\ &\le C_b J. \end{aligned} \tag{C13}$$

Finally $R_m \le CJ/r(m)$, whence

$$\frac{1}{d} \sum_m R_m = O(\log(2 + J)) = o(J). \tag{C14}$$

For fixed $0 < \delta < 1$, the columns with $|m| > (1-\delta)J$ and their incident edges thus contribute at most $C_b(\delta + J^{-1})$ to $U_j[b]/J$.

2. The interior limit

Fix $\delta > 0$ and restrict to $x_m = m/J$ with $|x_m| \le 1 - \delta$. All errors below are uniform over this set; constants may depend on b and δ. Put $f(x) = \sqrt{1 - x^2}$. Centered expansion gives $k_m - B_m = O_\delta(J^{-1})$. A global remainder estimate is

$$\begin{aligned} &\left| q(m + z) - B_m + \frac{m}{B_m} z \right| \\ &\quad \le \left(\frac{1}{2B_m} + \frac{4m^2}{B_m^3} \right) z^2 + \frac{z^4}{B_m^3}. \end{aligned} \tag{C15}$$

This follows from $|\sqrt{u_+} - B - (u - B^2)/(2B)| \le (u - B^2)^2/(2B^3)$, valid for every real u and $B > 0$, by taking $u = J^2 - (m + z)^2$. Thus the estimate also covers the region where q vanishes. Set $z = \sqrt{Ja(x_m)}Y$. Its right-hand side is $O_{b,\delta}(Y^2 + J^{-1}Y^4)$. Squaring and using the Gaussian moments through order eight gives

$$\begin{aligned} \frac{I_m}{J} &= a(x_m) f'(x_m)^2 + O_{b,\delta}(J^{-1/2}) \\ &= x_m^2 b(x_m) + O_{b,\delta}(J^{-1/2}). \end{aligned} \tag{C16}$$

For the gradient term, use the weighted norm $\|w\|_q^2 = \int q(t)|w(t)|^2 dt$. In an interior unit interval, $S \asymp_{b,\delta} J$, $S' = O_b(1)$, and $S'' = O_b(J^{-1})$. Equation (C10) and one further derivative imply

$$\begin{aligned} \|\partial_s v_s\|_q &= O_{b,\delta}(1), \\ \|\partial_s^2 v_s\|_q &= O_{b,\delta}(J^{-1/2}). \end{aligned} \tag{C17}$$

Indeed, $\partial_s^2 v_s / v_s$ is a polynomial in Y of degree at most four, with coefficients $O_{b,\delta}(J^{-1})$, and $q \le J$. Two applications of the fundamental theorem of calculus and Minkowski's inequality give

$$\begin{aligned} &\|v_{m+1} - v_m - \partial_m v_m\|_q \\ &\quad \le \int_m^{m+1} \int_m^s \|\partial_u^2 v_u\|_q du\, ds \\ &\quad = O_{b,\delta}(J^{-1/2}). \end{aligned} \tag{C18}$$

Thus the squared norm of $v_{m+1} - v_m$ differs from that of $\partial_m v_m$ by $O_{b,\delta}(J^{-1/2})$. In (C10), the squared translation term contributes

$$\begin{aligned} &\frac{1}{4Ja(x_m)} \mathbb{E}\left[q(m + \sqrt{Ja(x_m)}Y) Y^2 \right] \\ &\quad = \frac{f(x_m)}{4a(x_m)} + O_{b,\delta}(J^{-1/2}). \end{aligned}$$

The cross term containing S' is $O_{b,\delta}(J^{-1/2})$ and its square is $O_{b,\delta}(J^{-1})$. Since $c_m/J = f(x_m) + O_\delta(J^{-1})$, we obtain

$$\frac{G_m}{J} = \frac{1}{4b(x_m)} + O_{b,\delta}(J^{-1/2}). \tag{C19}$$

The Riemann sums in (C16) and (C19), divided by $d = 2J$, give the integral in (C4) over the interior. Equations (C9), (C13), and (C14) allow $\delta \downarrow 0$ after $j \to \infty$. This proves (C4), since $J/j \to 1$.

3. One explicit width sequence

Pointwise minimization of the integrand in (C4) suggests $b_*(x) = 1/(2|x|)$ away from the equator. Use instead

$$\begin{aligned} b_\epsilon(x) &= \frac{1}{2\sqrt{x^2 + \epsilon^2}}, \\ a_\epsilon(x) &= \frac{1 - x^2}{2\sqrt{x^2 + \epsilon^2}}, \qquad 0 < \epsilon \le 1. \end{aligned} \tag{C20}$$

An exact integral evaluates the limiting functional:

$$\frac{1}{2}\int_{-1}^{1}\left(x^2 b_\epsilon+\frac{1}{4b_\epsilon}\right)dx=\frac{1}{2}\sqrt{1+\epsilon^2}. \tag{C21}$$

For fixed ϵ this already implies $\limsup_j V_j/j \le \sqrt{1+\epsilon^2}/2$. Together with Proposition 4, it proves (8). To specify a single attaining family, we retain the parameter dependence of the estimates above.

There is an absolute constant $C>0$ such that whenever $0<\delta<1/2$, $J\ge\epsilon^{-1}$, and $J\delta\ge 1$,

$$\begin{aligned}\left|\frac{U_j[b_\epsilon]}{J}-\frac{1}{2}\sqrt{1+\epsilon^2}\right| &\le C\epsilon^{-7}(\delta+J^{-1})\\ &\quad + C(\epsilon^{-2}\delta^{-6}+\epsilon^{-12}\delta^{-4})J^{-1/2}+C\frac{1+\log(2J)}{J}.\end{aligned} \tag{C22}$$

We give explicit bounds supporting these exponents. The formulas in (C20) give

$$\begin{aligned}&\frac{1}{2\sqrt{2}}\le b_\epsilon\le\frac{1}{2\epsilon},\\ &\|a_\epsilon'\|_\infty\le C\epsilon^{-3},\qquad \|a_\epsilon''\|_\infty\le C\epsilon^{-5}.\end{aligned} \tag{C23}$$

Equation (C8) then gives $I_m\le C\epsilon^{-2}J$. In (C11) one may take the constant $C\epsilon^{-1/2}$. Using $S\ge cr$ and $|S'|\le C\epsilon^{-3}$ in (C12) gives $G_m\le C\epsilon^{-7}J$. The deleted pole regions therefore produce the first term of (C22); the corresponding integral tails are at most $C\epsilon^{-1}\delta$.

In the interior, $S\ge cJ\delta$, $S\le CJ\epsilon^{-1}$, and S', JS'' have the polynomial bounds (C23). To check all derivative constants, write the bracket in (C10) as $g=AY+B(Y^2-1)$, where $A=(2\sqrt{S})^{-1}$ and $B=S'/(4S)$. Then $\partial_s^2 v_s/v_s=g^2+\partial_s g$, and

$$\begin{aligned}\partial_s Y&=-S^{-1/2}-\frac{S'}{2S}Y,\qquad A'=-\frac{S'}{4S^{3/2}},\\ B'&=\frac{S''}{4S}-\frac{S'^2}{4S^2}.\end{aligned}$$

The four types of coefficient in $\partial_s^2 v_s/v_s$ are S^{-1}, $S'/S^{3/2}$, S''/S, and S'^2/S^2. Hence, using $q\le J$ and fixed Gaussian moments,

$$\begin{aligned}&\|\partial_s v_s\|_q\le C\epsilon^{-3}\delta^{-1},\\ &\|\partial_s^2 v_s\|_q\le C\epsilon^{-6}\delta^{-2}J^{-1/2}.\end{aligned} \tag{C24}$$

The finite-difference replacement therefore changes the squared norm by at most $C\epsilon^{-12}\delta^{-4}J^{-1/2}$. For the position term, use the global remainder (C15), $S\le CJ\epsilon^{-1}$, $B_m\ge J\sqrt{\delta}$, and $|k_m-B_m|\le CJ^{-1}\delta^{-3/2}$. Gaussian moments through order eight give the uniform bound

$$\left|I_m/J-x_m^2 b_\epsilon(x_m)\right|\le C\epsilon^{-2}\delta^{-6}J^{-1/2}. \tag{C25}$$

For the gradient expectation one can use the global bound (C7) directly:

$$\begin{aligned}&|q(m+\sqrt{S}Y)-B_m|\\ &\quad\le C\left[\sqrt{J}\epsilon^{-1/2}\delta^{-1/2}|Y|+\epsilon^{-1}\delta^{-1/2}Y^2\right].\end{aligned}$$

The errors in the squared translation term, the cross term with S', and its square are at most $C\epsilon^{-1}\delta^{-2}J^{-1/2}$, $C\epsilon^{-3}\delta^{-2}J^{-1/2}$, and $C\epsilon^{-6}\delta^{-2}J^{-1}$, respectively. Including (C24) and the change from c_m/J to $f(x_m)$,

$$|G_m/J-1/(4b_\epsilon(x_m))|\le C\epsilon^{-12}\delta^{-4}J^{-1/2}. \tag{C26}$$

Finally the derivative of $x^2 b_\epsilon+1/(4b_\epsilon)$ has absolute value at most $3/2$, uniformly in ϵ. This controls the Riemann-sum error. These are all terms in the preceding argument and prove (C22).

Now use (10) and, only in the proof, take $\delta_j=\epsilon_j^8$. The pole error is $O(1/\log J)+O((\log J)^7/J)$. The interior error is $O((\log J)^{50}/\sqrt{J})$. Every term tends to zero. Therefore $U_j[b_{\epsilon_j}]/j\to 1/2$. Since $L_j\le V_j\le W_j[b_{\epsilon_j}]\le U_j[b_{\epsilon_j}]$, (C5) follows. The cutoff δ_j has no role in the measurement; all columns and all real outcomes t remain present.

Appendix D: Schur averaging and completion of the main proof

Write $\mathbb{E}_N$ for expectation under $P_{N,j}$. The passage from spin asymptotics to copy asymptotics can be made without exchanging an uncontrolled limit and a growing sum.

Lemma 6. *The Schur probabilities satisfy*

$$\begin{aligned}\mathbb{E}_N[j(j+1)]&=\frac{3N}{4},\\ \frac{\mathbb{E}_N j}{\sqrt{N}}&\longrightarrow\sqrt{\frac{2}{\pi}}.\end{aligned} \tag{D1}$$

If a fixed sequence r_j, defined on all nonnegative integer and half-integer spins, has finite entries and $r_j=o(j)$, then $\mathbb{E}_N r_j=o(\sqrt{N})$.

Proof. The first identity is the trace of $(J_x^{\mathrm{tot}})^2+(J_y^{\mathrm{tot}})^2+(J_z^{\mathrm{tot}})^2$ in $I/2^N$. Let S_N be a sum of N independent symmetric signs. Summation by parts in $P_{N,j}=(2j+1)(q_j-q_{j+1})$, $q_j=2^{-N}\binom{N}{N/2-j}$, gives the exact identity

$$\mathbb{E}_N j=\mathbb{E}|S_N|-\frac{1}{2}+\frac{1}{2}\Pr(S_N=0). \tag{D2}$$

The binomial sums are

$$\mathbb{E}|S_{2n}|=2n\frac{\binom{2n}{n}}{2^{2n}},\qquad \mathbb{E}|S_{2n+1}|=(2n+1)\frac{\binom{2n}{n}}{2^{2n}}.$$

Stirling's formula proves (D1) for both parities. For any $\eta>0$, choose a fixed K_η such that $|r_j|\le\eta j$ when $j\ge K_\eta$. If $M_\eta=\max_{j<K_\eta}|r_j|$, then

$$\frac{\mathbb{E}_N|r_j|}{\sqrt{N}}\le\eta\frac{\mathbb{E}_N j}{\sqrt{N}}+\frac{M_\eta}{\sqrt{N}}.$$

Letting first $N\to\infty$ and then $\eta\downarrow 0$ proves the claim. In particular, finitely many small-spin values cause no problem. □

By Theorem 5 and Lemma 6,

$$\mathbb{E}_N W_j^{\mathrm{G}} = \frac{1}{2}\sqrt{\frac{2}{\pi}}\sqrt{N} + o(\sqrt{N}). \tag{D3}$$

The same conclusion holds with W_j^{G} replaced by either V_j or L_j. For any of these three sequences Z_j,

$$\begin{aligned}\frac{4N}{2N - 4\mathbb{E}_N Z_j} &= 2 + \frac{4\mathbb{E}_N Z_j}{N} + o(N^{-1/2}) \\ &= 2 + \sqrt{\frac{8}{\pi}} N^{-1/2} + o(N^{-1/2}).\end{aligned} \tag{D4}$$

The lower bound holds for every POVM by Proposition 4, and the explicit family supplies the matching upper bound. This completes the proof of Theorem 2.

Appendix E: The NH formula and strict finite-copy separation

Here we supply the finite-dimensional facts used in Section V. The two-parameter NH bound can be expressed as a convex first-moment optimization, or equivalently as its usual semidefinite relaxation [1, 2]. Put $D = 2^N$ and $Z = X + iY$ for arbitrary Hermitian first moments. Eliminating the auxiliary operator blocks gives the objective

$$\begin{aligned}\mathcal{F}(Z) &= \min_{W \succeq Z^\dagger Z,\, W \succeq ZZ^\dagger} \operatorname{tr} W \\ &= \operatorname{tr} Z^\dagger Z + \max_{0 \preceq R \preceq I} \operatorname{tr} R(ZZ^\dagger - Z^\dagger Z).\end{aligned}$$

The equality follows by taking the positive spectral part of $ZZ^\dagger - Z^\dagger Z$. The normalized NH risk is (N/D) times the minimum of $\mathcal{F}$ under the first-moment constraints.

Use the Schur basis with each multiplicity label retained, and let R be diagonal with the entries r_i from Appendix B on each spin block. Then

$$\mathcal{F}(Z) \geq \sum_{\alpha,\beta} (1 + r_\alpha - r_\beta)|Z_{\alpha\beta}|^2.$$

Every coefficient is nonnegative. On a raising edge it is $1 - r_i + r_{i+1} = c_i/a_i$. The unbiasedness constraints imply $\sum_{\text{raising edges}} c_i \operatorname{Re} Z_{i+1,i} = D$. Thus Cauchy–Schwarz on these edges gives $\mathcal{F}(Z) \geq D^2/S_{\mathrm{tot}}$, where $S_{\mathrm{tot}} = \sum_j \nu_j S_j$. This bound applies to all complex Z, including entries connecting different spin blocks.

To attain it, take adjacent raising matrices in each sector with real nonnegative entries $x_{j,i}$, identical on the multiplicity space. Set $w_j = P_{N,j}/d$. Their objective before multiplication by N is

$$\begin{aligned}&\sum_j w_j f_j(x_j), \\ f_j(x) &= \sum_i x_i^2 + \frac{1}{2}\sum_{i=0}^{d-1} |x_i^2 - x_{i-1}^2| \\ &= \sum_{i=0}^{d-1} \max(x_{i-1}^2, x_i^2).\end{aligned} \tag{E1}$$

where $x_{-1} = x_{d-1} = 0$. The zero means and cross derivatives vanish; the remaining unbiasedness constraint is $\sum_j w_j \sum_i c_i x_{j,i} = 1$. Expressions containing S_j^{-1} below include only $j > 0$ sectors; the $j = 0$ sector contributes zero. With r_i defined in Appendix B,

$$\begin{aligned}f_j(x) &\geq \sum_i (1 - r_i + r_{i+1}) x_i^2 \\ &= \sum_i \frac{c_i}{a_i} x_i^2 \geq \frac{(\sum_i c_i x_i)^2}{S_j}.\end{aligned} \tag{E2}$$

The first inequality supports each maximum by a convex combination of its two entries. At $x_i = a_i$, it is an equality: r_i selects the larger entry on each uncapped tail and mixes only two equal entries on the plateau. The second inequality is Cauchy–Schwarz and is also an equality there. A further weighted Cauchy–Schwarz inequality across sectors proves that the minimum is $1/\sum_j w_j S_j$, achieved by $x_{j,i} = a_{j,i}/\sum_k w_k S_k$. Multiplication by N proves (20).

Suppose now that the seed maximum (A2) attains the upper bound $4S_j/d$. Every seed then saturates the nonnegative terms of (B4) with $t = h/n \geq 0$. For $t > 0$, the kernel recurrence and connectedness force every coordinate of a nonzero seed to be positive. An edge with $c_i > \tau_j$ has $r_{i+1} > 0$ and $1 - r_i > 0$, so the two adjacent equations force $t = \tau_j$. All such seeds are therefore multiples of a single vector u. Its plateau coordinates can be set to one, and on the left tail

$$u_i = (a_i/\tau_j) u_{i+1}; \tag{E3}$$

the right tail is its reflection. For $t = 0$, the matrix in (B4) is diagonal and only its two endpoint entries can vanish; all interior λ_i are strictly positive.

If $c_1 < \tau_j$, the first two edges on the left are uncapped. Equation (E3) then forces every positive-t seed to satisfy $v_1 = (c_1/\tau_j)v_2$, while every zero-t seed vanishes at both nodes. Hence the sum of their effects obeys $M_{11} = (c_1/\tau_j)^2 M_{22}$, which is incompatible with $M = I$. Attainment of the seed maximum makes the resulting inequality strict.

To verify $c_1 < \tau_j$ for all $j \geq 5$, set $n = 2j \geq 10$. At $n = 10$ the six central edges give

$$\begin{aligned}\sum_i (c_i - c_1)_+ &= 2(\sqrt{24} + \sqrt{28} + \sqrt{30}) - 6\sqrt{18} \\ &> \sqrt{18} = c_1.\end{aligned}$$

For a fully rational check, the three ratios to $\sqrt{18}$ exceed respectively $23/20$, $6/5$, and $5/4$; twice their sum minus six exceeds one. For each fixed $i \geq 2$,

$$\frac{c_i}{c_1} = \sqrt{\frac{(i+1)(n-i)}{2(n-1)}}$$

increases with n. The three edges $i = 2, 3, 4$ and their three reflections remain distinct for $n \geq 10$. Thus $\sum_i (c_i -$

$c_1)_+ > c_1$ for every such n, and the monotonicity of (B1) implies $\tau_j > c_1$. For $N \geq 10$ the largest-spin sector $j = N/2$ has positive probability and a strict Fisher deficit, proving $C_N > C_N^{\mathrm{NH}}$.

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